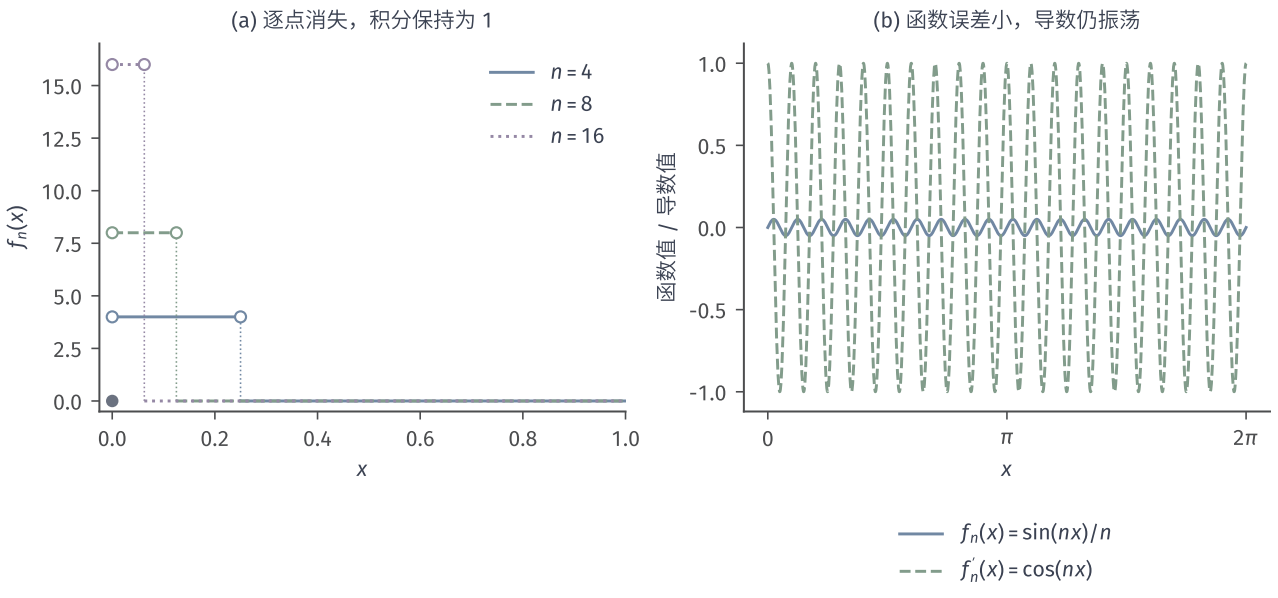
(a) 逐点消失，积分保持为 1
$f_n(x)$
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
x
$n = 4$
$n = 8$
$n = 16$
(b) 函数误差小，导数仍振荡
函数值 / 导数值
1.0
0.5
0.0
-0.5
-1.0
0
π
2π
x
$f_n(x) = \sin(nx)/n$
$f_n'(x) = \cos(nx)$